

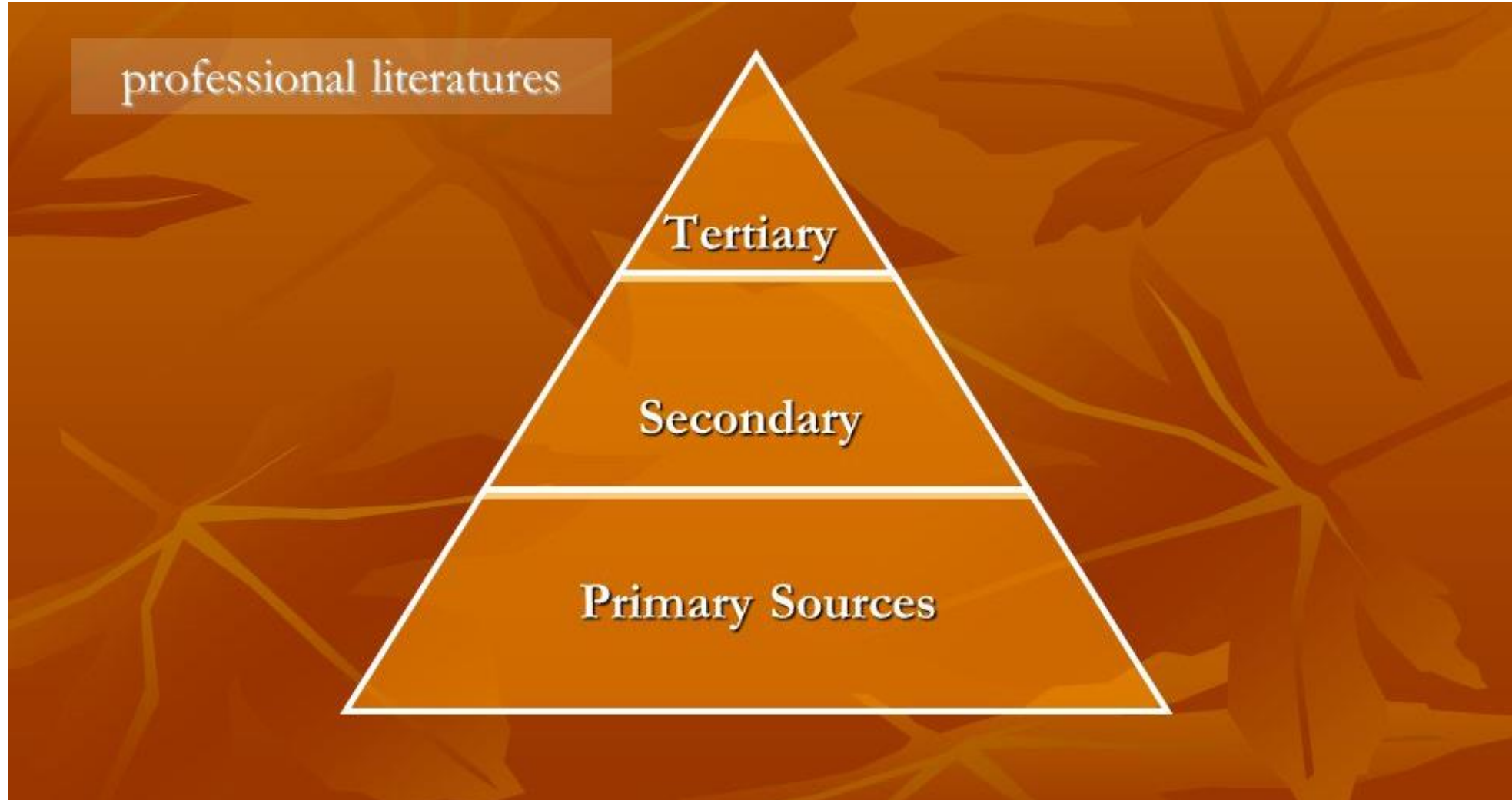
We can **define drug information** as a knowledge of facts through reading, study, or practice experience on chemical substance that is used in diagnosis, prevention and treatment of a disease.

- **Drug Information Service:**

It describes the activities undertaken by pharmacists in providing information to optimize drug use. It is those services which cover the activities of specially trained individual to provide accurate, unbiased, factual information in response to patient oriented drug problems that are received from health care professionals.

the best method to find information includes a stepwise approach moving first through **tertiary** (e.g. textbooks, full-text databases, and review articles) then **secondary** (e.g. indexing or abstracting service) and finally **primary** (e.g. clinical studies) literatures.

Drug information Sources :



Tertiary resources :

-provide information that has been summarized and distilled by the author or editor to provide a quick easy summary of a topic These include not only textbooks and compendia but also electronic databases (e.g., Micromedex, Lexicomp) and review articles.

-The best tertiary resources are written by experts in the field and are peer reviewed. If the tertiary resource is not current or comprehensive, a secondary resource should be consulted to locate primary literature on the topic. However, some questions can only be answered by using tertiary sources.

Advantages of Tertiary Resources:

These references may often serve as an initial place to identify information, due to:

1. They provide a fairly complete and concise overview of information available on a specific topic.
2. These resources are also convenient.
3. easy to use
- 4.familiar to most practitioners.

Disadvantages of Tertiary Resources:

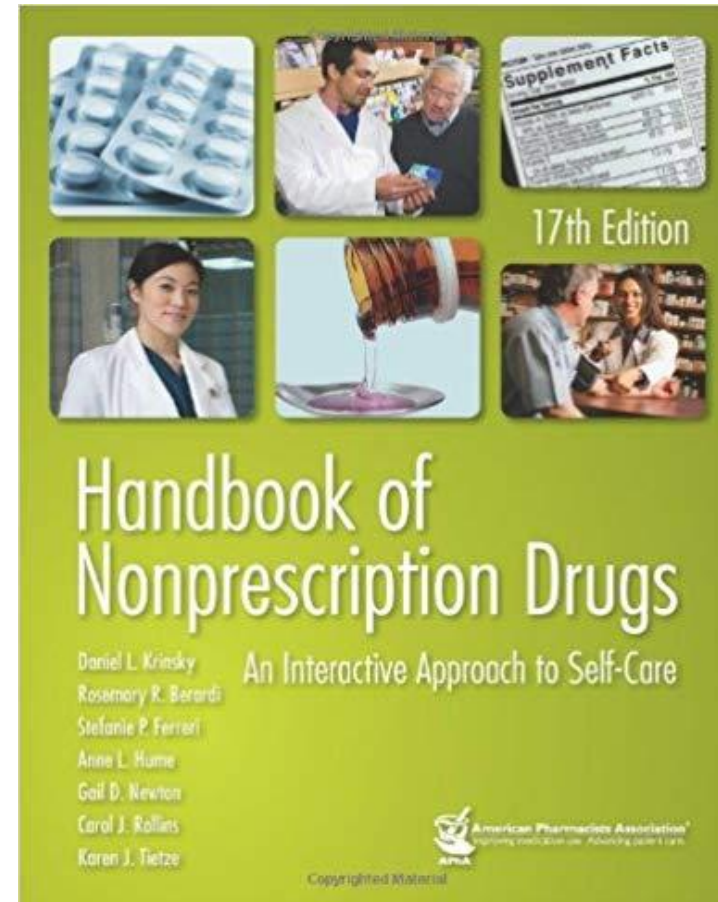
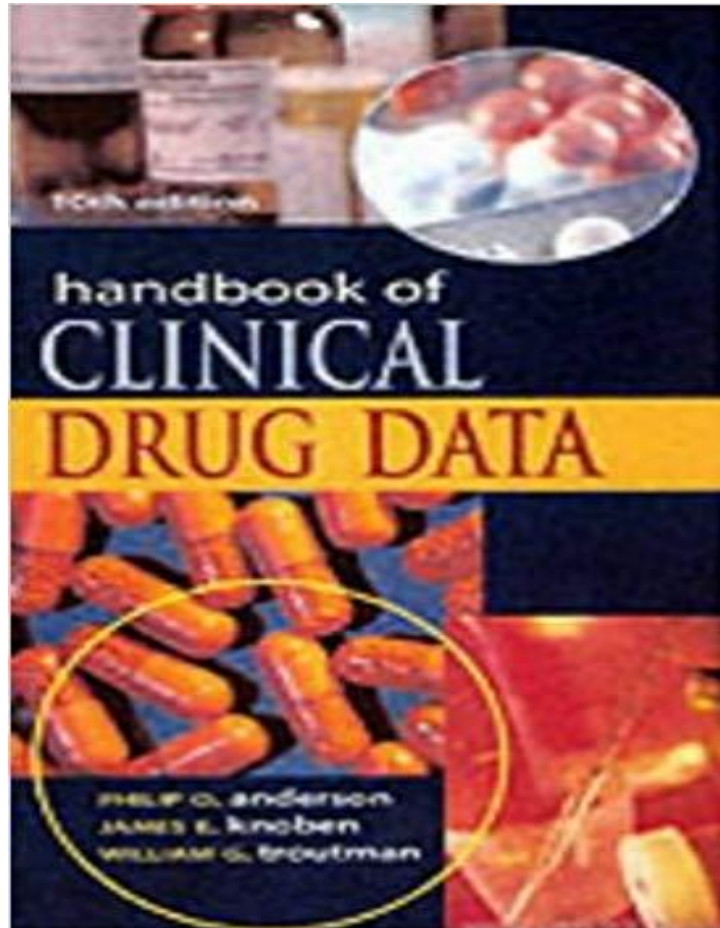
1. The lag time associated with publication of print copy tertiary resources, resulting in less current information.
2. The potential for author biases to be introduced
3. The fact that the information may be incomplete, misleading, or inaccurate .

Evaluation of Tertiary Literature

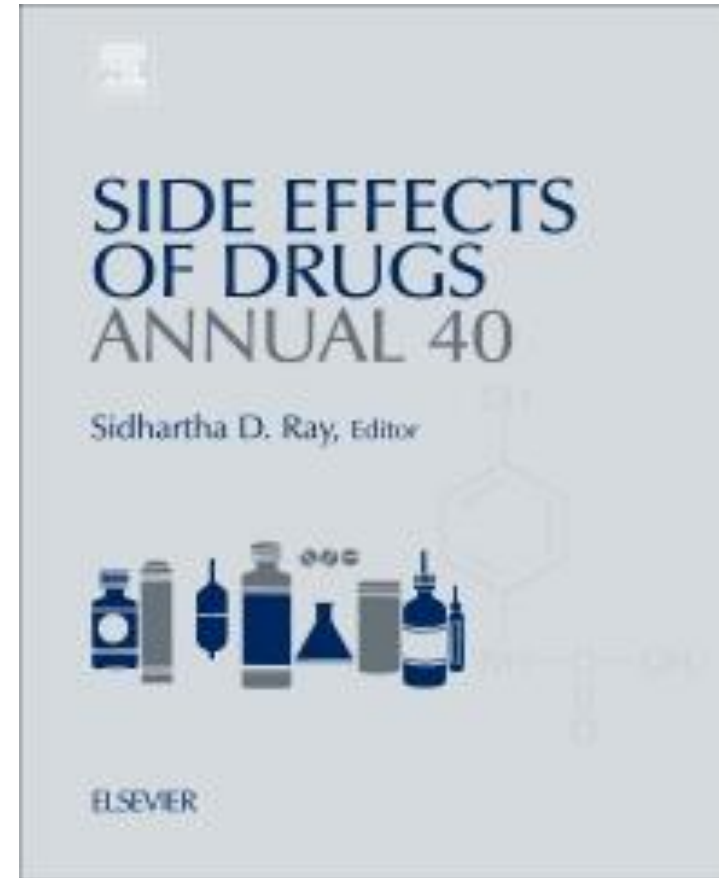
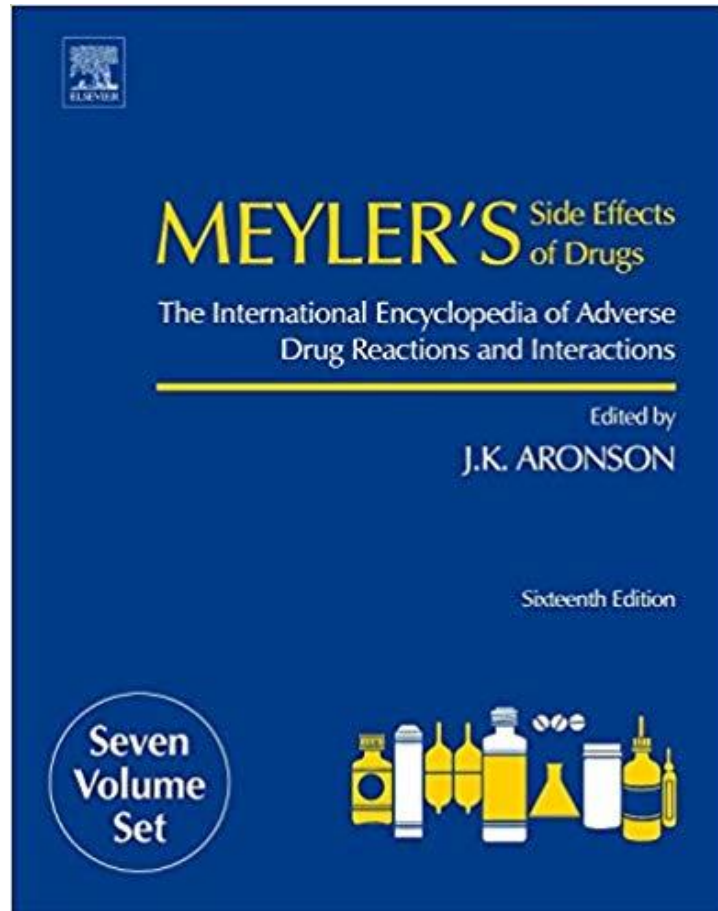
- 1- Does the author have appropriate •
experience/expertise to publish in this area? •
- 2- Is the information likely to be timely based on publication date?
- 3- Is the information supported with •
citations? •
- 4- Does the resource contain relevant information? •
- 5- Is the resource free of bias or errors? •
- 6- Is it clear/concise/easy-to-use ? •

*Example of tertiary drug sources :

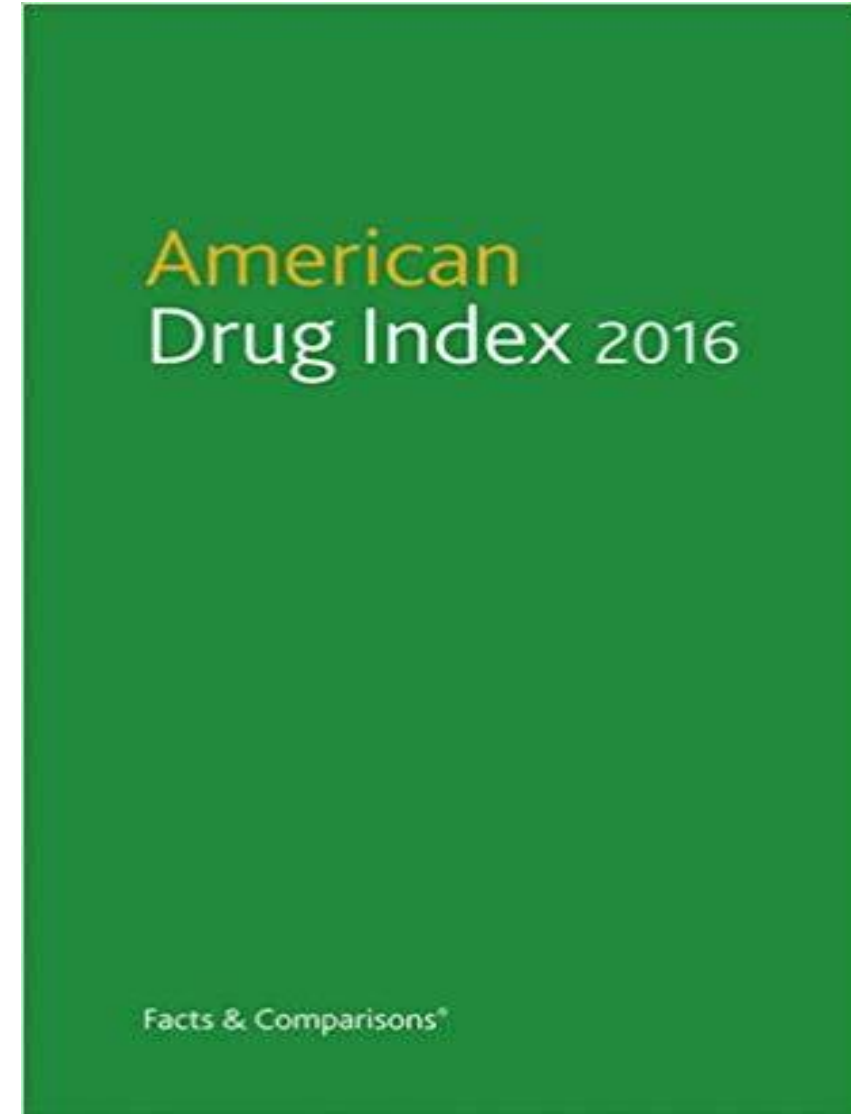
1-General product information



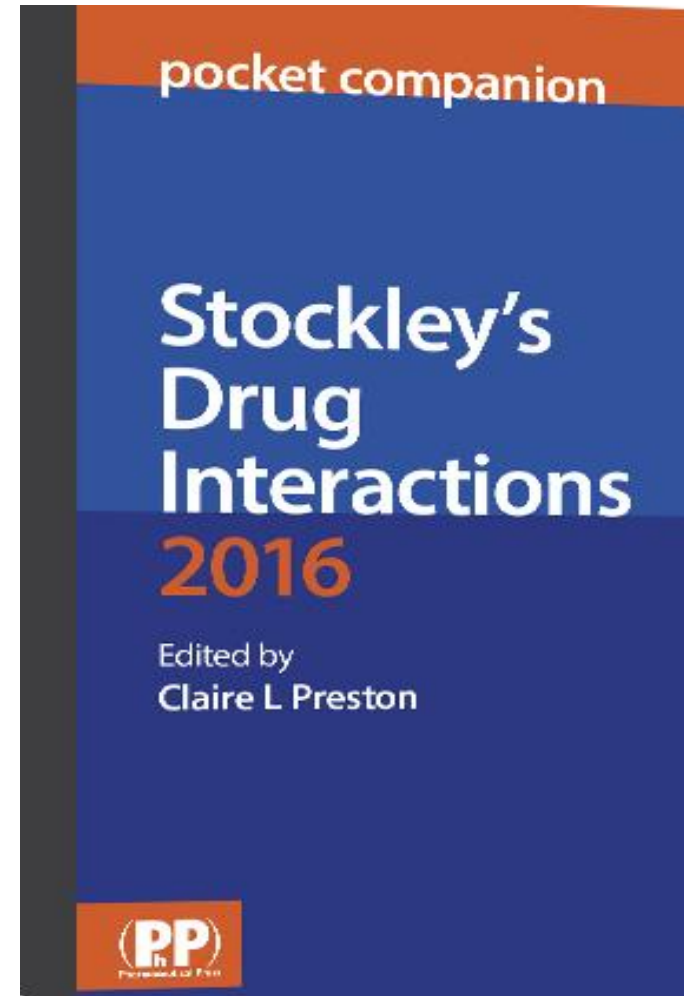
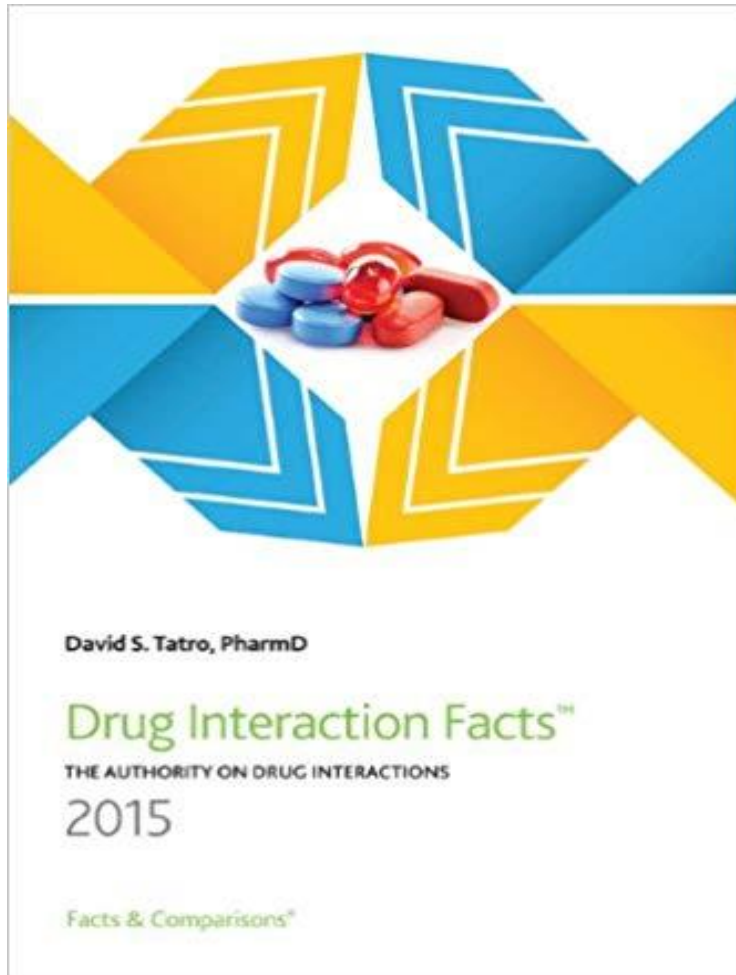
2-Adverse effects



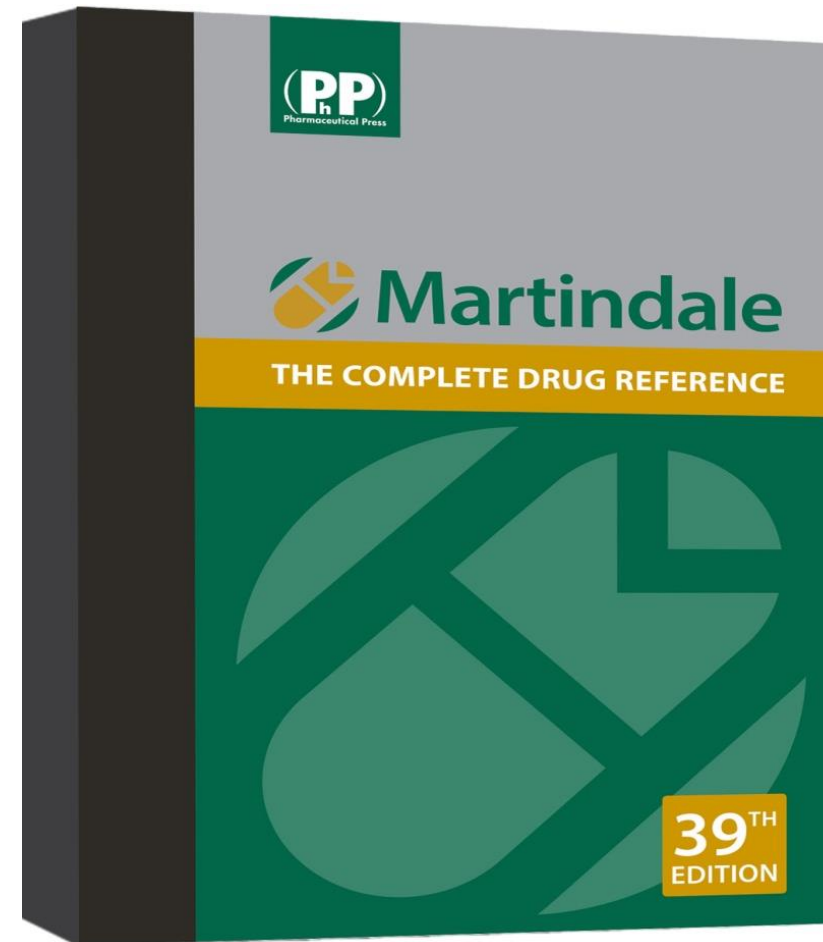
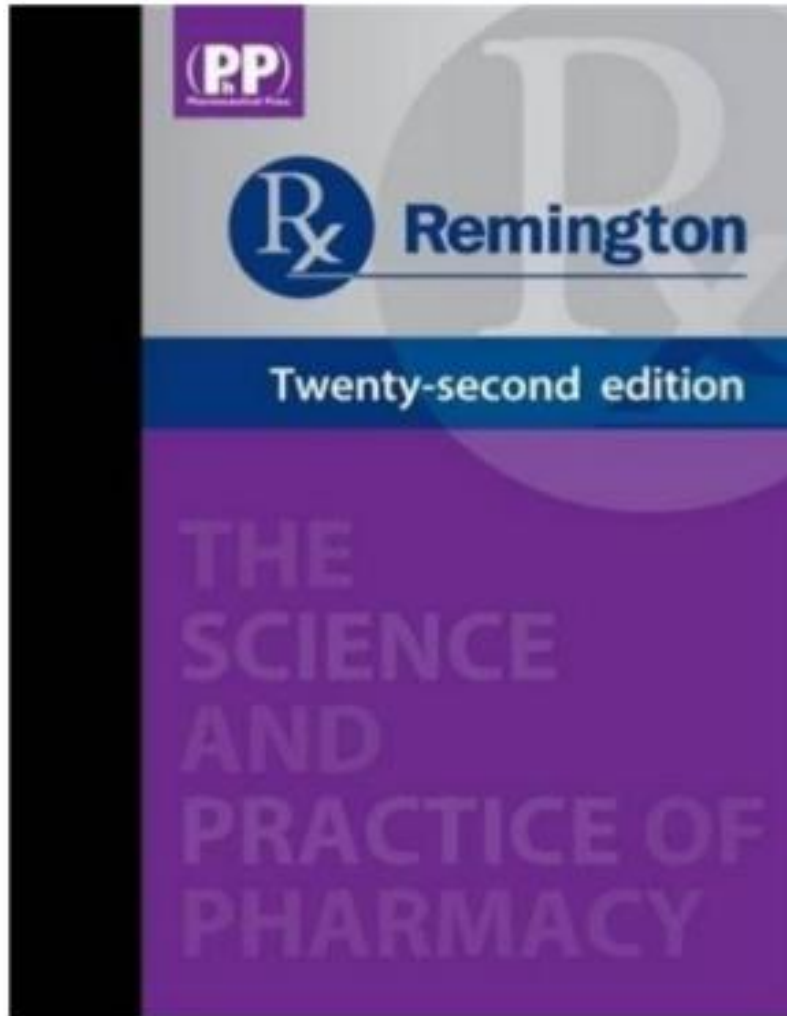
3-Availability of dosage form



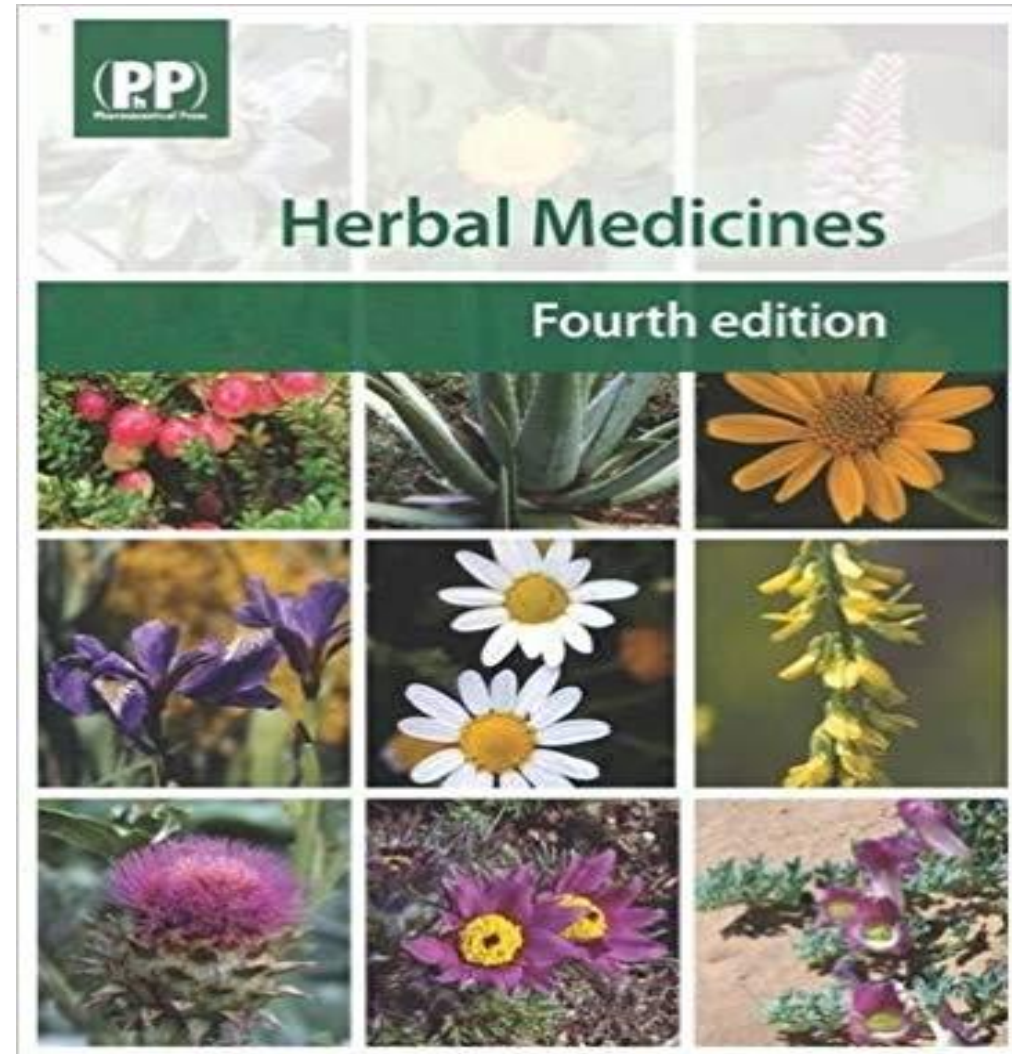
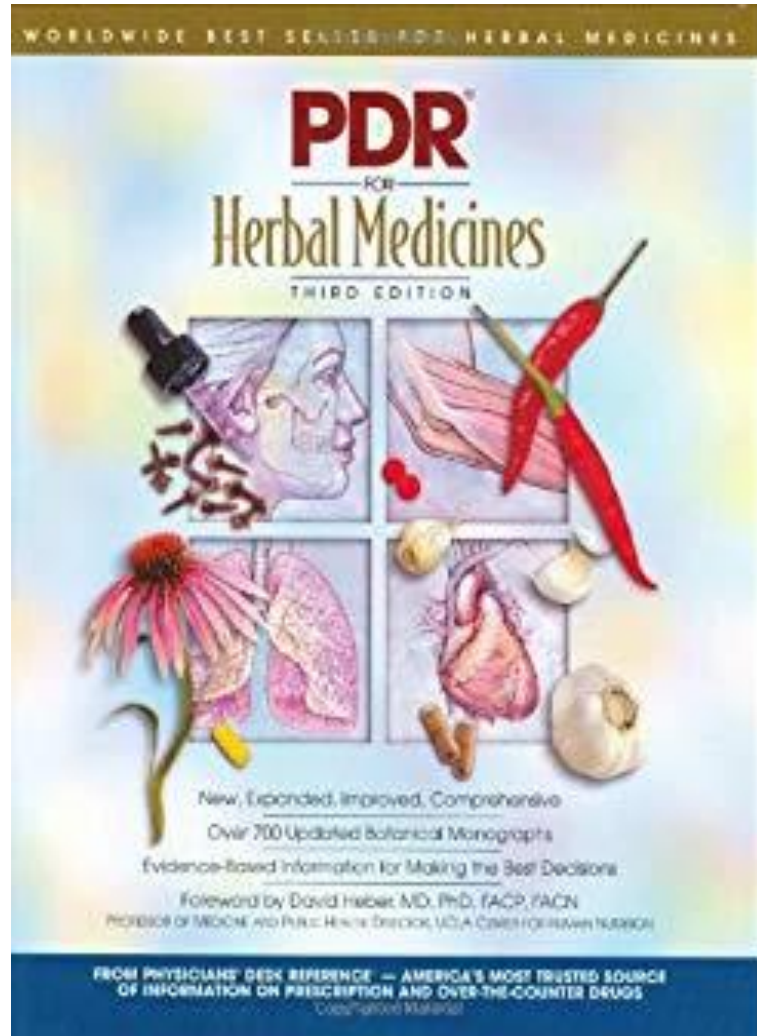
4-Drug interactions



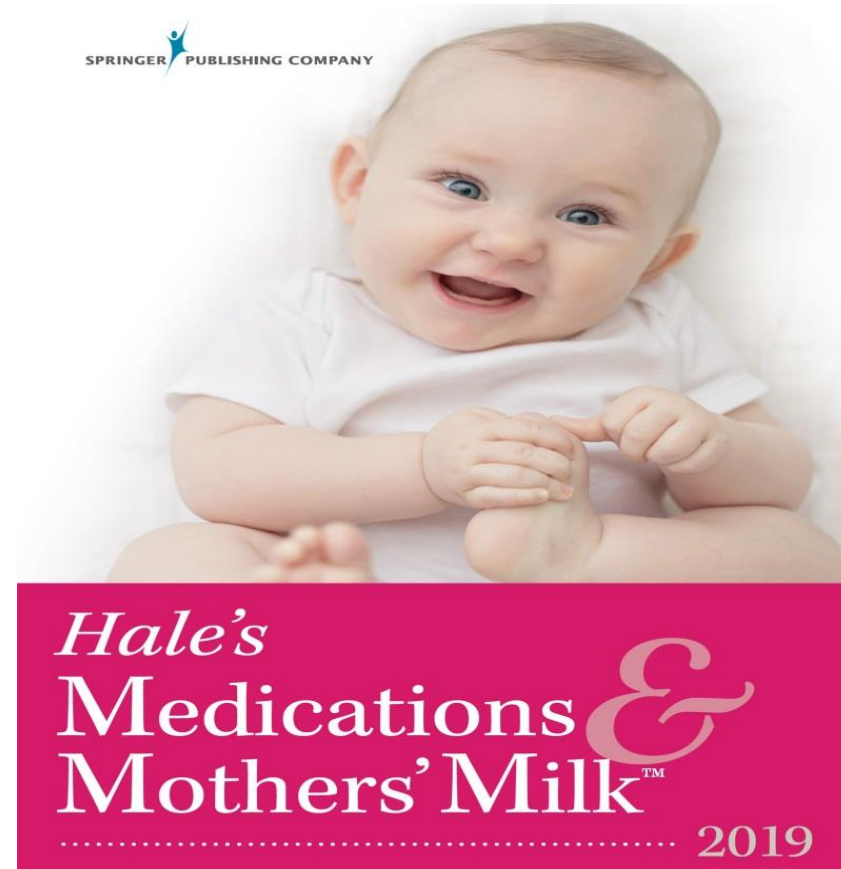
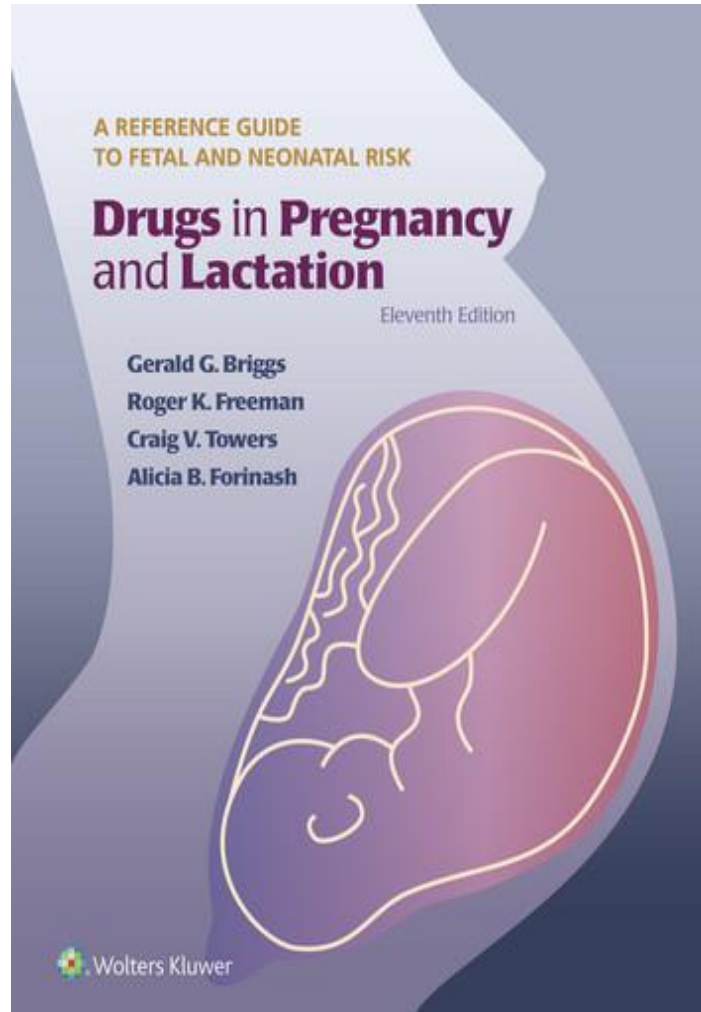
5-Formulations



6-Dietary supplements

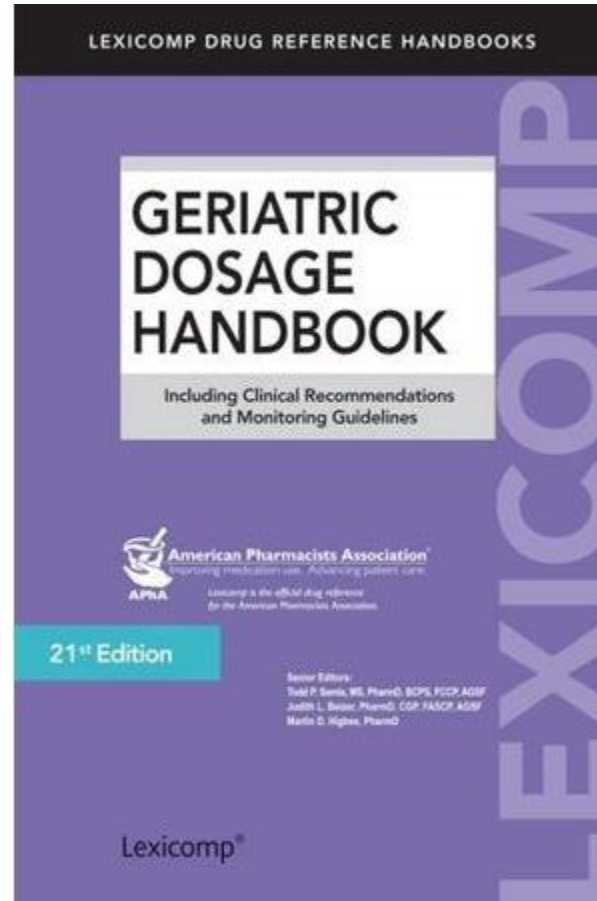


7-Drugs in pregnancy & lactation

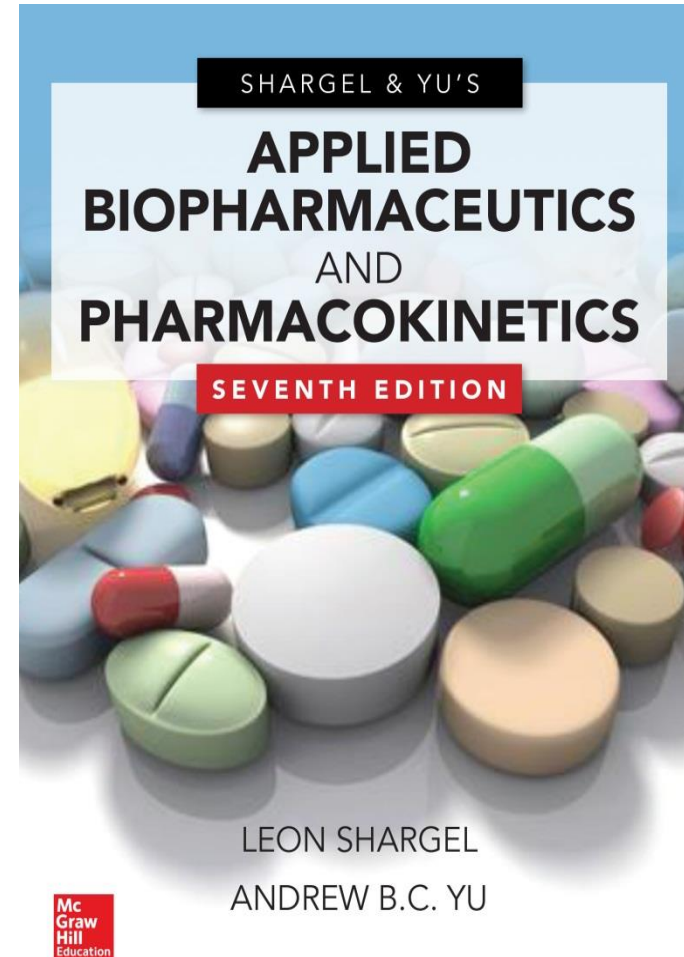
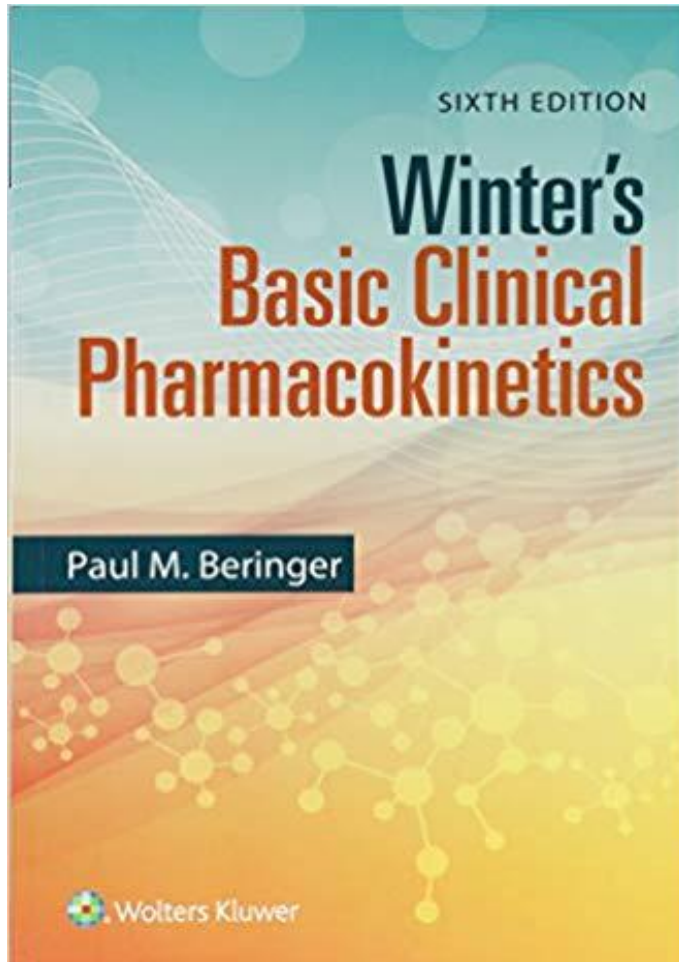


Thomas W. Hale, R.Ph., Ph.D.

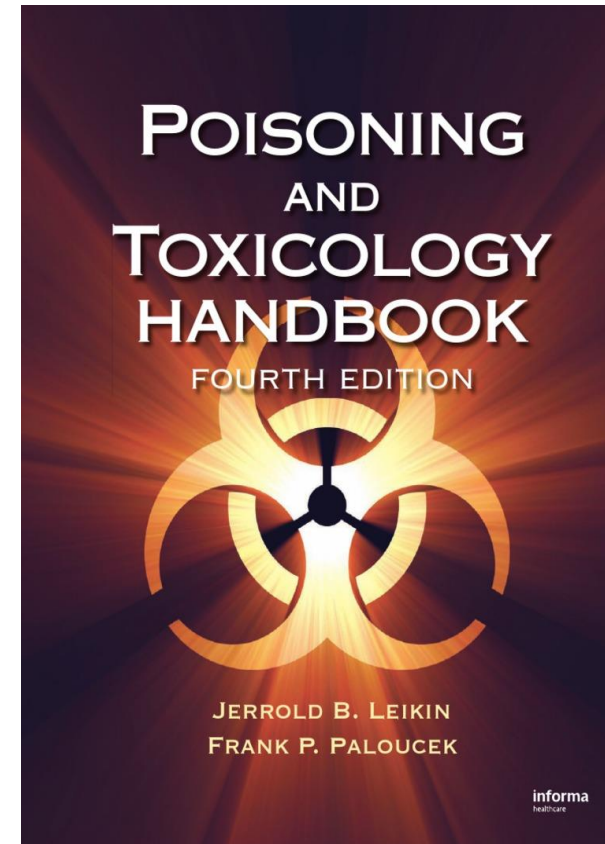
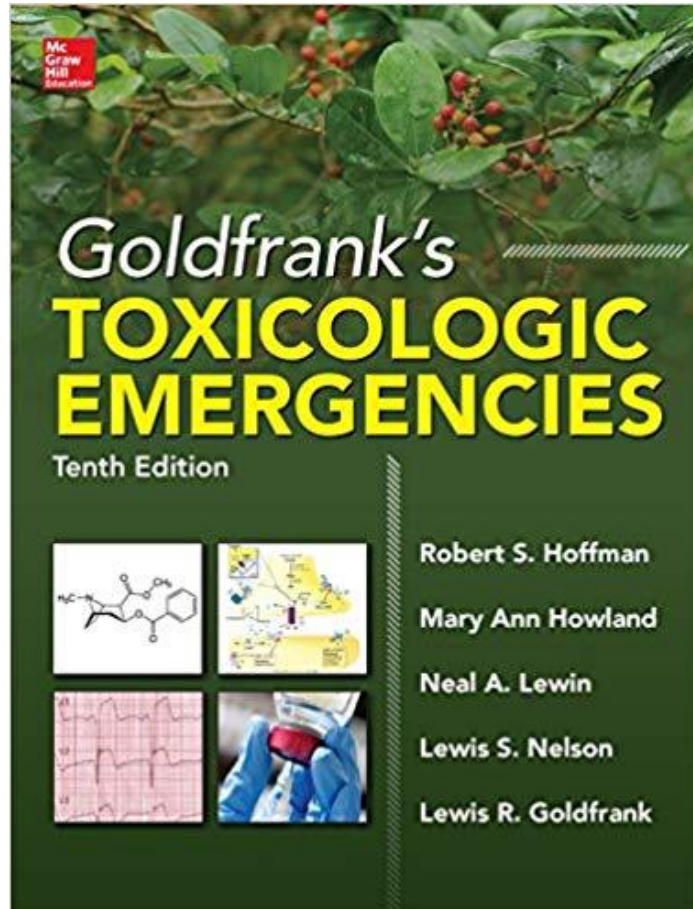
8-Geriatric dosage recommendations



9-Pharmacokinetics



10-Toxicology information



Electronic and Online tertiary information resources

Advantages of online tertiary information resources

- 1.They may be easier to use
- 2. Allow quicker access to information
- 3. Allow multiple searches to be performed simultaneously
- 4. Often contain the most recent information available regarding a topic

Medscape®

Medscape Registration

https://www.medscape.com/features/mktg/public/register/ppc1?src=ppc_bing

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Medscape

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 **Find accurate answers at the point-of-care**
Access the latest evidence based diagnosis and treatment information, and search over 15,000+ drugs, diseases, and procedures in our clinical reference.

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Go to Settings to activate Windows.

TOXNET from NIH

The screenshot shows the TOXNET website in a web browser. The browser's address bar displays <https://toxnet.nlm.nih.gov/>. The website header features the NIH logo and the text "U.S. National Library of Medicine" and "TOXNET TOXICOLOGY DATA NETWORK". Navigation links for "Mobile", "Help", "FAQs", "TOXNET Fact Sheet", and "Training Manual & Schedule" are present. A large molecular structure image is on the left, and a "Share" button is on the right. The main heading "Welcome to TOXNET" is followed by the description: "Your resource for searching databases on toxicology, hazardous chemicals, environmental health, and toxic releases". Below this is a search section with three tabs: "SEARCH TOXNET" (selected), "BROWSE", and "ADVANCED SEARCH". The "SEARCH TOXNET" tab contains a search input field with the placeholder text "e.g. benzene, endocrine disruptor", a dropdown menu labeled "ALL DATABASES", and a "Search" button. To the right of the search section is a "Environmental Health & Toxicology" sidebar with a "Visit Site" button. At the bottom, the "TOXNET Databases" section lists "MOST VISITED BY TOXNET USERS": "HSDB" (Hazardous Substances Data Bank) and "TOXLINE" (4 million references to literature on biochemical, pharmacological, physiological, and toxicological effects of drugs and other chemicals). A "Did you know" section at the bottom right provides a tip about choosing a database. A Windows watermark is visible in the bottom right corner of the browser window.

TOXNET TOXICOLOGY DATA NETWORK

Mobile | Help | FAQs | TOXNET Fact Sheet | Training Manual & Schedule

Welcome to TOXNET

Your resource for searching databases on toxicology, hazardous chemicals, environmental health, and toxic releases

SEARCH TOXNET Search all or select specific databases

BROWSE

ADVANCED SEARCH

e.g. benzene, endocrine disruptor

ALL DATABASES

Search

Environmental Health & Toxicology

Resources on environmental health and toxicology

Visit Site

TOXNET Databases

MOST VISITED BY TOXNET USERS

HSDB

Hazardous Substances Data Bank. Peer-reviewed toxicology data for over 5,000 hazardous chemicals

TOXLINE

4 million references to literature on biochemical, pharmacological, physiological, and toxicological effects of drugs and other chemicals

Did you know

There is a guide to choosing a database

Which Resource Should I Use

can help you pick the right resource for your search.

Activate Windows

Go to Settings to activate Windows.

Medline from NIH

The screenshot shows the MedlinePlus website in a web browser. The browser's address bar displays <https://medlineplus.gov/>. The website's header features the NIH logo and the text "U.S. National Library of Medicine". Below this is the MedlinePlus logo, which includes a stylized green and blue icon and the text "MedlinePlus® Trusted Health Information for You". A search bar with the placeholder text "Search MedlinePlus" and a green "GO" button is located to the right of the logo. Below the search bar are links for "About MedlinePlus", "Site Map", "FAQs", and "Customer Support". A navigation bar contains links for "Health Topics", "Drugs & Supplements", "Videos & Tools", and a language option "Español". The main content area is divided into several sections. On the left, there are three green boxes with white icons: an apple for "Health Topics" (Find information on health, wellness, disorders and conditions), a pill for "Drugs & Supplements" (Learn about prescription drugs, over-the-counter medicines, herbs, and supplements), and a play button for "Videos & Tools" (Discover tutorials, health and surgery videos, games, and quizzes). Below these is a section for "Lab Test Information" with a flask icon (Learn why your doctor orders laboratory tests and what the results may mean). On the right, there is a "Share MedlinePlus" section with icons for email, Facebook, and Twitter. Below this is a featured article titled "Check out our Creamy Squash Soup with Shredded Apples recipe." with a photo of a bowl of soup and text describing it as a quick-to-fix soup bursting with warm-you-up-flavor. Below the article are five small numbered boxes (1, 2, 3, 4, 5) and a play button icon. Further down, there is a "Tweets by @MedlinePlus" section showing a tweet from MedlinePlus.gov about Acute Flaccid Myelitis (AFM). To the right of the tweets is a "Stay Connected" section with a sign-up form for the "My MedlinePlus newsletter" and a green "GO" button. At the bottom right, there is a "NIH MedlinePlus" logo and a small image of a person. A watermark "Activate Windows" is visible in the bottom right corner of the browser window.

MedlinePlus - Health In

<https://medlineplus.gov/>

NIH U.S. National Library of Medicine

MedlinePlus®
Trusted Health Information for You

Search MedlinePlus GO

About MedlinePlus Site Map FAQs Customer Support

Health Topics Drugs & Supplements Videos & Tools Español

Health Topics
Find information on health, wellness, disorders and conditions

Drugs & Supplements
Learn about prescription drugs, over-the-counter medicines, herbs, and supplements

Videos & Tools
Discover tutorials, health and surgery videos, games, and quizzes

Lab Test Information
Learn why your doctor orders laboratory tests and what the results may mean

Share MedlinePlus

Check out our **Creamy Squash Soup with Shredded Apples** recipe.
This quick-to-fix soup is bursting with warm-you-up-flavor.

1 2 3 4 5

Tweets by @MedlinePlus

MedlinePlus.gov
@medlineplus

Acute Flaccid Myelitis (AFM) is in the news with the CDC investigating an increase of cases. What symptoms should you be on the look out for? [ow.ly/N6bn30mh0vA](https://www.cdc.gov/media/releases/2014/s0916-afm.html)

Stay Connected
Sign up for the My MedlinePlus newsletter

Enter email address GO

NIH MedlinePlus

FDA website

The screenshot shows a web browser window with multiple tabs. The active tab is titled "Information by Drug Class" and shows the URL <https://www.fda.gov/Drugs/DrugSafety/InformationbyDrugClass/ucm165107.htm>. The browser's address bar and tabs are visible at the top. The website header includes the U.S. Department of Health and Human Services logo, the FDA logo, and the text "U.S. FOOD & DRUG ADMINISTRATION". A search bar labeled "Search FDA" is on the right. Below the header is a navigation menu with links: Home, Food, Drugs, Medical Devices, Radiation-Emitting Products, Vaccines, Blood & Biologics, Animal & Veterinary, Cosmetics, and Tobacco Products. The main content area is titled "Drugs" and shows a breadcrumb trail: Home > Drugs > Drug Safety and Availability > Information by Drug Class. A blue button labeled "Information by Drug Class" is present. The main heading is "Acetaminophen Information". Below this heading are social media sharing buttons for Facebook, Twitter, LinkedIn, Pinterest, Email, and Print. The text describes Acetaminophen as an active ingredient in hundreds of over-the-counter (OTC) and prescription medicines, used to relieve pain and fever. It also mentions that in prescription medicines, it is often combined with other active ingredients to treat moderate to severe pain. A section titled "Pain Medicine Use During Pregnancy" contains a link to an FDA Drug Safety Communication dated 1/9/2015. On the left side, under "Resources for You", there are several links: "Questions and Answers: FDA warns of rare but serious skin reactions with the pain reliever/fever reducer acetaminophen", "Consumer Education: Using Acetaminophen and Nonsteroidal Anti-inflammatory Drugs Safely", "Sometimes Drugs and the Liver Don't Mix", and "A Guide to Safe Use of Pain Medicine". A watermark "Activate Windows" is visible in the bottom right corner.

U.S. Department of Health and Human Services

FDA U.S. FOOD & DRUG ADMINISTRATION

A to Z Index | Follow FDA | En Español

Search FDA

Home Food Drugs Medical Devices Radiation-Emitting Products Vaccines, Blood & Biologics Animal & Veterinary Cosmetics Tobacco Products

Drugs

Home > Drugs > Drug Safety and Availability > Information by Drug Class

Information by Drug Class

Acetaminophen Information

SHARE TWEET LINKEDIN PIN IT EMAIL PRINT

Acetaminophen is an active ingredient in hundreds of over-the-counter (OTC) and prescription medicines. It relieves pain and fever. And, it is also combined with other active ingredients in medicines that treat allergy, cough, colds, flu, and sleeplessness. In prescription medicines, acetaminophen is found with other active ingredients to treat moderate to severe pain. Acetaminophen can cause serious liver damage if more than directed is used. The FDA has taken action to improve the safety of consumers when using acetaminophen.

Pain Medicine Use During Pregnancy

- FDA Drug Safety Communication: FDA has reviewed possible risks of pain medicine use during pregnancy 1/9/2015

Resources for You

- Questions and Answers: FDA warns of rare but serious skin reactions with the pain reliever/fever reducer acetaminophen
- Consumer Education: Using Acetaminophen and Nonsteroidal Anti-inflammatory Drugs Safely
- Sometimes Drugs and the Liver Don't Mix
- A Guide to Safe Use of Pain Medicine

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Daily Med (side effects)

The screenshot shows a web browser window with three tabs: "FDA Drugs", "DailyMed", and "FDA Orange Book: Approved Dr". The address bar shows the URL "https://dailymed.nlm.nih.gov/dailymed/". The website header includes the NIH logo and "U.S. NATIONAL LIBRARY OF MEDICINE". A navigation bar contains links for "REPORT ADVERSE EVENTS" and "RECALLS". The main content area features the "DAILYMED" logo and a search bar with the placeholder text "Enter drug, NDC code, drug class, or Set ID". Below the search bar are tabs for "ALL DRUGS", "HUMAN DRUGS", and "ANIMAL DRUGS". A section titled "MORE WAYS TO SEARCH:" includes links for "ADVANCED SEARCH", "BROWSE DRUG CLASSES", "LABEL ARCHIVES", and "TABLET/CAPSULE ID TOOL". A message states: "This website contains 105556 drug listings as submitted to the Food and Drug Administration (FDA). At the present time, this Web site does not contain a complete listing of labels for approved prescription drugs." Below this are two columns of content. The left column is titled "NEWS" and contains a link to "DailyMed Announcements", a post date of "December 19, 2017", and a section titled "Drug Listing Certification" with text about the FDA deadline for drug listings. The right column is titled "FDA GUIDANCES & INFORMATION" and contains a link to "Drug Guidance, Compliance & Regulatory Information", an FDA logo, and links to "View FDA Structured Product Labeling Resources", "View FDA Drug Labeling Guidances", and "View All FDA Drug Guidances". At the bottom right, there is a watermark that says "Activate Windows Go to Settings to activate Windows."

NIH U.S. NATIONAL LIBRARY OF MEDICINE

REPORT ADVERSE EVENTS | RECALLS

DAILYMED

ALL DRUGS | HUMAN DRUGS | ANIMAL DRUGS

Enter drug, NDC code, drug class, or Set ID

MORE WAYS TO SEARCH: ADVANCED SEARCH BROWSE DRUG CLASSES LABEL ARCHIVES TABLET/CAPSULE ID TOOL

This website contains 105556 drug listings as submitted to the Food and Drug Administration (FDA).
At the present time, this Web site does not contain a complete listing of labels for approved prescription drugs.

SHARE

NEWS

[DailyMed Announcements](#)

Posted: December 19, 2017

Drug Listing Certification

The U.S. Food and Drug Administration is reminding the pharmaceutical industry of the December 31, 2017, deadline to update or certify their drug listings with FDA. This applies to drug listings that were not initially listed or updated during the current calendar year. This is the first deadline of the annual certification requirement under

FDA GUIDANCES & INFORMATION

[Drug Guidance, Compliance & Regulatory Information](#)

 [View FDA Structured Product Labeling Resources](#)
[View FDA Drug Labeling Guidances](#)
[View All FDA Drug Guidances](#)

NLM SPL RESOURCES

The following Structured Product Labeling (SPL) resources have

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 **Pill Identifier Tool** >
Quick, Easy, Pill Identification

 **Drug Interaction Tool** >
Check Potential Drug Interactions

 **Pharmacy Locator Tool** >
Including 24 Hour, Pharmacies

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Secondary resources

- include indexing and abstracting systems that organize and provide easy retrieval of primary resources.
- Indexing systems include the article citation, with or without access to the abstract; some include a link to the full-text article.
- Abstracting systems provide not only the citation but also the abstract and often a link to the full-text article.

Advantages:

- Simple search strategies
- Very current citation information
- Access point for tremendous amount of primary sources

Disadvantages:

- Often require more expertise to use than primary or tertiary resources.
- Retrieved references must be filtered for quality.
- Pharmacists should obtain and evaluate the original article because abstracts may not include enough information.
- Too complex for patients

PubMed Central

The screenshot shows the PubMed Central website. At the top, the browser address bar displays the URL <https://www.ncbi.nlm.nih.gov/pmc/>. Below the address bar, the PMC logo is visible, along with a dropdown menu currently set to 'PMC' and a search bar. The main content area features a large banner with the PMC logo and a description: 'PubMed Central® (PMC) is a free full-text archive of biomedical and life sciences journal literature at the U.S. National Institutes of Health's National Library of Medicine (NIH/NLM)'. To the right of the banner is a 'PubReader' section. Below the banner, there are three columns of links. The left column, titled 'Get Started', includes links for 'PMC Overview', 'Users' Guide' (highlighted with a red box), 'Journal List', 'PMC FAQs', and 'PMC Copyright Notice'. The middle column, titled 'Participate', includes links for 'Information for...', 'Add a Journal', 'Participation...', 'File Submission', and 'File Validation'. The right column, titled 'Public Access', includes links for 'Funders and PMC', 'How Papers Get Into PMC', 'NIH Manuscript Submission System', 'My Bibliography', and 'PMCID/PMID/NIHMSID Converter'. A large text box with a red border and a light blue background is overlaid on the right side of the page, containing the following text: 'PubMed Central (PMC) is an archive of articles that have been deposited in this free repository. Over 4.8 million articles are available. Access is via journal A-Z listing and keyword searches for articles are entered into the PMC search box.'

Get Started

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- [Users' Guide](#)
- [Journal List](#)
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Other Resources

- [PMC International](#)
- [Text Mining Collections](#)
- [Developer Resources](#)
- [NLM LitArch](#)
- [PMC Citation Search](#)

4.8 MILLION Articles
are archived in PMC.
Content provided in part by:

2110	330	4623
Full Participation Journals	NIH Portfolio Journals	Selective Deposit





☒ Articles (☒ include patents) ☐ Case law

New! 2014 Scholar Metrics released

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